



Friedrich-Alexander-Universität
Rechts- und Wirtschafts-
wissenschaftliche Fakultät

*Interpreting Multi-Word Expressions in Text-to-
Image Generation: A Cross-Attention Approach
Using Stable Diffusion*

Master's Thesis

Handed in by:	Sayef Ali, Khan
Matriculation number:	22929049
Program of study:	Information and Communication Technology
Professor:	Prof. Dr. Michael Amberg
Supervisor:	Dr. Stefan, Arnold
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Abstract

Recent advancements in text-to-image generation mostly through the latent diffusion model Stable Diffusion have greatly improved. Many achievements have been made, but they still cannot get the subtleties and contextual interpretations of multi-word expressions. MWEs, or multi-word expressions, often contain complex meaning. As a result, they pose a special challenge for generative models to connect text and image meaningfully. The study looks forward to understand these complexities by adapting techniques such as Directed Attention Attribution Mechanism (DAAM) to a large MWE repository. DAAM helps create two-dimensional attribution maps which show how words and phrases affect the spatial and semantic properties of generated images.

This study uses Intersection over Union (IoU) as a metric to assess the extent to which the holistic or segmented meaning of MWEs aligns with its picture. The methodology integrates Linguistics and advanced Generative Artificial Intelligence techniques. The tool offers a framework for interpreting how diffusion models process complex textual inputs. This work advances the broader mission of enhancing generative Artificial Intelligence systems, creating circumstances for the development of systems capable of generating context-rich and, at times, semantically ambiguous textual data that are interpretable for us.

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List of abbreviations

DAAM	Diffusion Attention Attribution Maps
MWE	Multi-Word Expression
IoU	Intersection over Union
LDM	Latent Diffusion Model
AI	Artificial Intelligence
GAN	Generative Adversarial Networks
VAE	Variational Autoencoders
DDPM	Denoising Diffusion Probabilistic Model
Grad-CAM	Gradient-weighted Class Activation Mapping
LIME	Local Interpretable Model-Agnostic Explanations
CNN	Convolutional Neural Network
FID	Fréchet Inception Distance
CLIP	Contrastive Language-Image Pretraining
MWE-CWI	Multiword Expression Complexity Word Identification

1. Introduction

The ability of machines to create coherent images from text is a major leap forward in artificial intelligence, known as text-to-image generation. Openlaender et al. (2023) illustrate how this has revolutionized fields such as healthcare, education and entertainment, amongst many others. The success of Stable diffusion and other latent diffusion models is increasing. These models are now setting the standard for the creation of photorealistic and contextual images. Models use complex computational structures—most notably cross-attention systems—which enable precise correspondence between text prompts and their visual analogs (Zhang et al., 2024).

Even though a lot of progress has been made, it is still very hard to understand and describe multi-word expressions (MWEs). MWEs are dialectal structures for example, idioms, collocations, and domain-specific phrases that are hard to take literally. Phrases, like “spill the beans” or “chronic fatigue,” consist of deeper and larger meanings than the sum of their parts (Miletić & Walde, 2024). Such situational and semantically difficult nature makes it hard for generative systems to balance the subtleness of language with the beauty of images.

The fact that MWEs are so important in many areas shows how useful they are. In entertainment, they help tell stories by making them more appealing to the consumers. In education, they help students understand metaphors and learn about other cultures. In technical areas like healthcare, it's important to correctly read MWEs in medical texts to avoid misunderstandings that could be life-threatening (Contreras Kallens & Christiansen, 2022). The complexity of MWEs goes beyond dialectical complexities and represents the need for practical necessity for AI systems to create meaningful visual outputs.

1.1 Motivation for the Study

The biggest motivation for this study is derived from the underlying complex issues offered by MWEs as well as the distinctive opportunities provided by the Multiword Expression Complexity Word Identification dataset (MWE-CWI) (Ekochmar, n.d. 2020). With AI systems gaining grip in significant fields, it has become increasingly essential to ensure semantic and cultural accuracy in interpretability, where text-to-image jobs call for higher precision (Tang et al., 2022a). This is especially concerning in applications needing high semantic precision, such as medical imaging and instructional content development.

The Multiword Expression Complexity Word Identification dataset (MWE-CWI) (Ekochmar, n.d. 2020) has substantially advanced this research. This rigorously annotated dataset contains varied MWEs with varying levels of complexity and probabilistic interpretations, providing a solid foundation for assessing and evaluating MWE interpretability. The dataset, which includes annotations such as “Complex binary,” “Complex probabilistic,” and contextual markers, provides a rich testing ground for evaluating how latent diffusion models such as Stable Diffusion handle linguistic complexities. Its comprehensiveness provides a wide range of circumstances, making it ideal for furthering our understanding of how MWEs influence text-to-image generation.

By treating MWEs as both a language and conceptual issue for AI, the present work tackles the interpretability issues that are at the heart of our research questions. As we move ahead with establishing how DAAM (Tang et al., 2022a) might shed light on model attention and how a particular word influences the sum of the MWE interpretation, the dataset and the problem framing ensure future studies are clearly linked to these fundamental investigation goals.

1.2 Addressing Interpretability through DAAM

Latent diffusion models’ reliance on cross-attention mechanisms has made possible the dynamic association of textual elements with visual features; a process essential for MWE representation (Miletič & Walde, 2024). However, the ambiguity of these systems makes it more problematic to assess how well models understand and depict difficult language (Liu et al., 2024). A tool such as Diffusion Attention Attribution Maps DAAM (Tang et al., 2022a) offer a possible answer to this problem. Two-dimensional visual overlays produced by DAAM show how specific words or phrases affect several components of the produced image. This study uses DAAM with the Multiword Expression Complexity Word Identification dataset (MWE-CWI) (Ekochmar, n.d. 2020) to investigate the link between language structures and visual outputs, therefore clarifying the interpretability of MWEs in generative artificial intelligence.

From a more general standpoint, MWEs are ubiquitous in communication, trespassing languages and fields; they are not limited to a single linguistic or cultural setting. AI systems meant to operate in several language and cultural environments depend on exact understanding of MWEs. While studies by Tang et al. (2022a) demonstrate their relevance in domain-specific applications, research by Contreras Kallens and Christiansen (2022) underlines the need of MWEs in enhancing cross-cultural

understanding. This universal character emphasizes the need for models able to provide realistic images considering the cultural and linguistic subtleties of MWEs.

1.3 Research Questions

This study investigates two important research problems to solve the above-mentioned difficulties and progress the knowledge of multi-word expressions (MWEs) in generative artificial intelligence.

First, to address a critical challenge in aligning language structures with visual outputs the study examines how Diffusion Attention Attribution Maps (DAAM) can be employed to envision model attention specific to MWEs during the process of text-to-image generation.

Second, it examines the influence of individual words within MWEs on their collective semantic interpretation in generative models. Directing the study with these questions in hand, the study aims to provide multi-layered insights into the interpretability of MWEs and the methods underlying their visual and semantic alignment in AI systems.

This paper consists of several sections outlining the theoretical background, methodology and finding followed by a depth discussion on the findings within the contexts of existing research.

Chapter 2 reviews the fundamentals of AI, outlining the main subfields of interest such as supervised and unsupervised learning. The development of generative models is traced, especially latent diffusion frameworks, and the linguistic challenges posed by multi-word expressions (MWEs). It further examines cross-attention mechanisms, bringing to light their key role in connecting textual prompts with corresponding visual representations. It also investigates visuolinguistic phenomena, putting much emphasis on the intersection of linguistic constructs with visual output. This shall be an investigation that will help develop a better understanding of how insights from linguistics and computer vision might help better design and work with generative models.

Chapter 3 fully describes the research design by elaborating on the selection and use of the Multiword Expression Complexity Word Identification dataset (MWE-CWI) (Ekochmar, n.d. 2020), the Stable Diffusion model, and the application of Diffusion Attention Attribution Maps (DAAM). It introduces quantitative evaluation metrics, including Intersection over Union (IoU), which measures the extent of alignment between the words in MWEs and their visual representations.

Focusing on the interpretation and representation of multi-word expressions (MWEs), Chapter 4, shows the results on the effectiveness of Stable Diffusion and DAAM. It presents important new perspectives on model attention and alignment supported by visual and statistical data, therefore providing a thorough analysis of the strengths and constraints of these generative models.

Chapter 5 places the results of the study in the framework of current research thereby stressing their relevance for the development of generative artificial intelligence systems. It also notes its constraints and looks at possible uses of the study.

The final chapter brings together the key contributions of this study and brings forth an outlook on future research directions and improvement for generative AI.

The interpretability challenges of MWEs, taken up in this thesis, represent a further contribution to the refinement of generative AI systems. The research bridges the gap between computational efficiency and semantic accuracy and lays the foundation for more reliable and culturally sensitive AI applications.

2. Theoretical background

This section explores the foundational theories and technologies relevant to understanding how MWEs are processed in text-to-image generative models, directly addressing the research question on semantic representation. Artificial Intelligence (AI) generative capabilities have evolved significantly, with Stable Diffusion representing a pivotal development in latent diffusion models. Building on the strengths of Generative Adversarial Networks (GANs) and Variational Autoencoders (VAEs), Stable Diffusion employs a latent-space-focused denoising process to produce high-resolution, semantically aligned visuals (Goodfellow et al., 2014; Rombach et al., 2022).

The strength of generative AI lies in its innovative architectures, which integrate machine learning with probabilistic reasoning. At the core of this thesis are diffusion models, particularly Stable Diffusion, which have set new benchmarks in text-to-image generation. Stable Diffusion, a latent diffusion model (LDM), combines probabilistic denoising techniques, latent space representations, and text embeddings to create semantically rich and coherent visuals from textual inputs. This chapter explores the theoretical and technical foundations of generative AI, focusing on the evolution of generative models, the challenges associated with multi-word expressions (MWEs), and

the methodologies and metrics employed to analyze and interpret these complex interactions.

2.1 Generative AI and Its Evolution

The field of Artificial intelligence (AI) has experienced revolutionary changes; generative artificial intelligence is now a major focus of study and application. Three main frameworks define generative models, meant to produce new data approximating input distributions: Generative Adversarial Networks (GANs), Variational Autoencoders (VAEs), and Diffusion Models. Each of these frameworks has its unique strengths, limitations, and applications.

Generative Adversarial Networks (GANs): GAN was first proposed by Goodfellow et al., in 2014; suggesting its reliance on a dual-network architecture: a generator and a discriminator. The former generates synthetic data, and the latter scores the authenticity of the data. This adversarial training drives successively better performance from the generator. Tasks like image synthesis, super-resolution, and style transfer have shown astonishing performance for GANs. For example, GANs power platforms like NVIDIA's GAN let users produce photorealistic graphics from basic sketches. GANs, however, are prone to training instability and problems like mode collapse, in which case the generator generates limited data variations, therefore failing to represent the whole data distribution (Salimans et al., 2016). These constraints limit their relevance in situations needing high diversity or strength.

Variational Autoencoders (VAEs): VAEs are generative models that address generative modeling through probabilistic latent variable learning, proposed by Kingma and Welling in 2014. VAEs encode data into a latent space and restructure it via a decoder. This enables smooth interpolation and representation learning. They have been effectively applied in anomaly detection, drug discovery, and latent space exploration. While being stable at training, these models produce blurry outputs particularly noticeable in image generation limiting their applications involving higher-resolution modalities.

Diffusion Models: Inspired by non-equilibrium thermodynamics, Diffusion models, generate data as a process of iterative denoising. Diffusion models generate outputs that approximate GANs in fidelity while avoiding their instability problems by injecting noise during training and learning to reverse it. Operating in a constrained latent space, a latent diffusion model called Stable Diffusion further maximizes this framework by lowering processing needs without compromising output quality (Rombach et al., 2022). These models have helped to enhance artificial intelligence-

assisted design and found uses in the creative sectors including media and art. Diffusion models' iterative character, however, raises computational complexity and thereby increases their resource-intensity relative to GANs and VAEs.

The development of these models emphasizes how increasingly sophisticated generative artificial intelligence is. Every model tackles a particular difficulty and has special benefits and restrictions. Knowing these subtleties helps one to improve applications like text-to-image generation, especially in linguistically difficult settings like multi-word expressions (MWEs).

2.2 Stable Diffusion: Architecture and Mechanisms

Stable Diffusion represents a state-of-the-art implementation of Latent Diffusion Models (LDMs), specifically designed for text-to-image generation. Its architecture, as depicted in Figure 1, integrates three core components: a Variational Autoencoder (VAE) for data compression, a denoising diffusion probabilistic model (DDPM) for iterative refinement, and advanced text encoders, such as CLIP and BERT, to ensure semantic alignment.

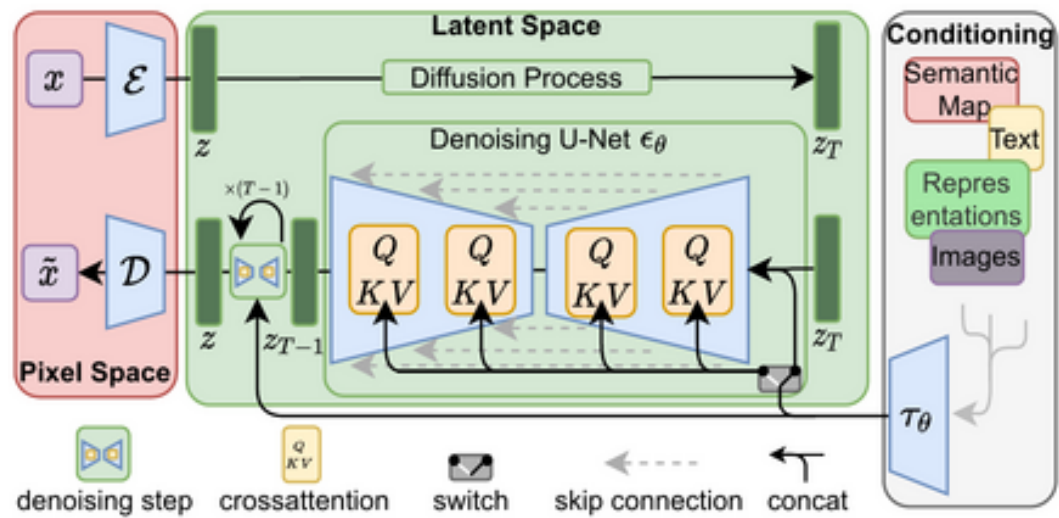


Figure 1: Detailed architecture of Stable Diffusion, illustrating the transition between pixel space and latent space, and the integration of conditioning inputs through cross-attention mechanisms (Rombach et al. 2022)

As shown in Figure 1, the workflow of Stable Diffusion can be divided into the following stages:

Pixel to Latent Compression via Variational Autoencoder (VAE): The encoder (E) starts the process by compressing the pixel data (x) into a latent representation

(z). This phase drastically lowers computing cost while preserving the semantic and structural features required for generation. During the last step, the decoder (D) reconstructs the image from the latent space, guaranteeing high-quality outputs with less processing time.

Denoising Diffusion Process in Latent Space: In latent space, a denoising diffusion process iteratively refines noisy latent representations into coherent images. The technique is done by utilizing a U-Net architecture (ϵ_θ) as indicated in Figure 1. The model gradually pushes the latent representation closer to the target data distribution using a series of denoising steps. Skip connections within the U-Net architecture are critical for retaining fine-grained spatial features across layers, resulting in high-resolution output.

Cross-Attention Mechanisms for Semantic Alignment: Stable Diffusion uses cross-attention techniques to dynamically align textual embeddings with spatial features in latent space. These processes compute attention ratings for each textual token (queries, (Q)) against spatial features (keys, (K)) and their corresponding values (V), directing the model to produce semantically consistent outputs.

Conditioning Inputs: The model includes conditioning inputs such as text embeddings, semantic mappings, and reference images, as seen on the right side of Figure 1. These inputs are incorporated into the diffusion process, allowing the model to produce contextually appropriate graphics adapted to a variety of applications and user needs.

Feature Integration Through Concatenation and Switching: To improve the generative process, Stable Diffusion employs advanced techniques such as feature concatenation and switching. These strategies make it easier to integrate intermediate characteristics, which improves the coherence and overall quality of the images produced.

Stable Diffusion 2.1 builds upon the core architecture of its predecessors, refining the process of text-to-image generation through enhanced efficiency and semantic alignment. Utilizing a latent diffusion framework, the model begins by compressing pixel data into a latent representation via a Variational Autoencoder (VAE), reducing computational overhead while preserving critical features (Rombach et al., 2022). In the latent space, a denoising diffusion process guided by a U-Net architecture iteratively refines noisy latent representations into coherent images. The integration of cross-attention mechanisms ensures that textual embeddings are dynamically aligned with spatial features, enabling the generation of semantically accurate visuals. Version 2.1 introduces improvements such as support for higher-resolution outputs,

refined prompt interpretation, and an expanded training dataset that enhances the model's ability to handle complex prompts and reduce biases (Stability AI, 2024).

Stable Diffusion 2.1 is available through multiple platforms, including open-source repositories such as Stability AI's GitHub page and popular AI model hosting services like Hugging Face. These platforms provide pre-trained models, documentation, and tools for fine-tuning, making it accessible to researchers, developers, and creative professionals.

2.3 MWEs in Computational Linguistics

Given their frequency and complexity in natural language, multi-word expressions (MWEs) are a fundamental topic of research in computational linguistics. MWEs—idioms, collocations, and compound nouns—represent language constructions whose meanings typically exceed the literal interpretation of their components. For instance, “spill the beans” suggests “revealing a secret,” a connotation not quite deducing from its individual phrases. For computational models, particularly generative artificial intelligence systems assigned with integrating verbal and visual representations, this semantic non-compositionality presents major difficulties. Cognitively speaking, MWEs are handled differently than single words or grammatical plain sentences. Studies show that idioms, including “kick the bucket,” are kept and accessed in the mental lexicon as lexicalized units (Cutting & Bock, 1997). This implies that models needing effective MWE representation must give contextual clues top priority over literal word meanings. Computationally, this corresponds with the need of attention mechanisms, including cross-attention in Stable Diffusion, to focus on the whole meaning of MWEs rather than their individual components.

According to linguistics, MWE's can be divided into syntactic and semantic. Important groups include Idiomatic Expressions are non-compositional expressions whose meaning is removed from its literal meaning (for instance, ‘to break the ice’). List of words that have a known relationship, like strong coffee. Compound nouns are nouns that consist of two or more words, which require contextual disambiguation (e.g. travel document) (Figure 6). Phrasal verbs, meaning verbs used alongside a preposition or adverb, have a different meaning than their original form for example, “give up.

The diverse languages show how hard MWEs are for generative AI to do. To produce valuable outputs, models must scan for idioms, grammatical differences, and context conditioned syntax. In machine translation, where literal translations often fall short to reflect their intended meaning, MWEs have long been a priority. To handle this, statistical and neural machine translation systems include phrase-level alignments and

attention techniques (Baldwin & Kim, 2010). In text-to-image generation, too, models such as Stable Diffusion map MWEs to suitable visual elements by using cross-attention. But as this thesis shows, generative models may struggle with idiomatic MWEs, therefore exposing a semantic understanding gap.

2.4 Cross-Attention and Visual-Linguistic Alignment

Interpretability is an essential component of generative AI research, offering knowledge about complicated models' decision-making processes and instilling trust in their results. As generative models get more complex, knowing how they convert inputs into outputs is crucial for enhancing their design and reliability. In this thesis, Diffusion Attention Attribution Maps (DAAM) were used to examine the interpretability of Stable Diffusion. However, it is critical to place DAAM within the larger landscape of interpretability tools by comparing it to other frequently used methods such as Gradient-weighted Class Activation Mapping (Grad-CAM) and Local Interpretable Model-Agnostic Explanations (LIME). These comparisons emphasize DAAM's distinct contributions while also emphasizing its benefits and limits.

Introduced by Selvaraju et al. (2017), Gradient-weighted Class Activation Mapping (Grad-CAM) is a widely employed method for convolutional neural network (CNNs) interpretation especially in image classification tasks. Grad-CAM emphasizes spatial areas in a picture that most help a given prediction by using the gradients entering the last convolutionary layer. For example, Grad-CAM can show in a model classifying dog breeds whether the network emphasizes identifying characteristics like the ears or coat patterns.

The strength of Grad-CAM is its capacity to provide visually instructive class-specific heatmaps explaining predictions. Its reliance on class-specific gradients, however, limits its applicability in multi-modal and generative models, where outputs originate from intricate interactions between textual and visual modalities rather than are connected to a single class. Grad-CAM lacks effectiveness in models such as Stable Diffusion, which uses cross-attention mechanisms to dynamically align linguistic and visual elements, since it cannot sufficiently depict the multi-dimensional connections inherent in such systems.

Proposed by Ribeiro et al. (2016), Local Interpretable Model-Agnostic Explanations (LIME) provide a flexible framework for understanding predictions by modifying input features and calculating their impact on outputs. From image recognition to text classification, LIME is appropriate for a broad spectrum of applications since it runs apart from the fundamental model architecture. LIME, for example, can identify in an

image classification model which areas of an image are most important for a given prediction by occluding or altering certain areas and tracking model confidence changes.

LIME is versatile; however, it struggles in high-dimensional data situations as cross-attention heatmaps produced by text-to-image models. LIME finds it challenging to create meaningful perturbations without oversimplifying the behavior of these systems since their features are simply highly complicated and interdependent. Moreover, although perceptive for individual predictions, LIME's localized explanations lack the global viewpoint required to grasp systematic trends over a dataset.

Designed especially for generative models such as Stable Diffusion, DAAM outperforms both Grad-CAM and LIME in handling the interpretability issues raised by multi-modal systems. DAAM illustrates the cross-attention distributions supporting text-to-image generation unlike Grad-CAM, which emphasizes uni-modal data. This enables DAAM to record intricate mappings between textual tokens and visual elements, therefore providing a more sophisticated knowledge of how generative models process multi-word expressions (MWEs). DAAM can show, for instance, whether Stable Diffusion breaks out “hard pressed” into its component parts or views it. DAAM distinguishes itself from LIME’s perturbation-based method also in its capacity to aggregate attention weights across several layers and timesteps. While LIME offers localized insights into individual predictions, DAAM presents a whole picture of attention distributions, that helps researchers to find consistent trends in model behavior. Analyzing complicated linguistic structures, where knowledge of the interaction between words and their contextual meanings is essential can be benefited especially from this.

Although DAAM thrives in storing multi-modal interactions, in some situations merging insights from Grad-CAM and LIME could improve interpretability even more. For example, Grad-CAM’s class-specific heatmaps may accentuate how particular elements affect textual token alignment, therefore complementing DAAM. Likewise, LIME’s model-agnostic explanations could offer other angles on feature importance, particularly in situations when DAAM heatmaps expose contrasting attention distributions. Combining these methods could allow researchers to have a more complete knowledge of generative model behavior, thereby bridging the uni-modal and multi-modal interpretability constraints.

2.5 Evaluation Metrics for Text-to-Image Models

Assessing the quality and interpretability of text-to-image models requires robust evaluation metrics. Key metrics include:

Fréchet Inception Distance (FID): An extensively used system of measurement for evaluating the quality of images generated by artificial intelligence (AI) models is Fréchet Inception Distance (FID). It is particularly useful for generative models like Generative Adversarial Networks (GANs) and diffusion models. The fundamentality of FID relies on the measurement of how similar the generated images are to real images, equally in terms of visual quality and the distribution of their characteristics. But how does this work, and why is it important? Well, imagine trying to compare two photo albums. One holding a picture taken by a professional photographer, the real images, and another produced by an AI model, the generated images. You should not only look at the pictures but also study their patterns, colors, textures, and other visual details to see how close they are to the real ones. This is basically what FID does. It checks how similar the pictures made by AI are to real ones. You want to see how similar the AI's photos are to the real ones, not just by looking at them but by analyzing patterns, colors, textures, and other visual details. This is essentially what FID does—it evaluates how close the AI-generated images are to real images.

FID employs a pretrained image classification model—usually the Inception network—that has already been trained to recognize and categorize images to accomplish this. FID Extracts Features. FID examines features of photos rather than plain pixels. The Inception model detects abstract patterns or qualities such as forms, edges, or textures. FID interprets the characteristics of generated and actual images as points in a mathematical space. It then computes these points' distributions for both sets of photos. To find the distance between these distributions, FID employs the Fréchet distance—a mathematical formula. The FID score will be low if the generated images resemble the genuine ones rather than being similar. A high FID score denotes that, in either quality or diversity, the generated images deviate greatly from the original ones.

Consider it as a comparison between the forms of two clouds of points in the sky. The FID score decreases with increasing overlapping of the clouds.

FID excels in spotting when generated images seem hazy or unrealistic. For instance, the FID score will rise representing the bad quality if an artificial intelligence model produces faces with deformed features. FID not only assesses personal image quality but also verifies whether the generated images are diversified. The FID score will be worse if an artificial intelligence model generates 100 identical faces than if a

collection of diverse and realistic faces. FID prevents from being deceived by surface similarities by examining feature distributions instead of raw pixel values. It detects in the images more subtle trends in line with human perception (Heusel et al., 2017).

FID's performance is linked to the model's strengths and limitations because it extracts features using the Inception network. If the Inception model has difficulty grasping certain sorts of images, FID scores may not appropriately reflect their quality. FID assesses visual similarity, but not semantic alignment. For example, if the AI provides a realistic image of a cat for a dog-related challenge, FID may still give a poor score because the image appears excellent despite being semantically inaccurate (Parmar et al., 2021). FID scores vary depending on the dataset utilized for comparison. Even if images are visually appealing or diverse, FID might falsely penalize generated images for having a particular style or feature distribution that differs from it (Borji, 2022).

FID is considerably used in research to benchmark the performance of generative models. Its efficiency in evaluating GANs is much more vigorous than prior metrics like Inception Score, as investigated by Heusel et al. (2017)

Mokady et al. (2021) employed FID to diffusion models, indicating how the metric captures improvements in image quality over prior methods. However, it is not free of errors in specific scenarios, such as when semantic alignment between prompts and generated images is critical (Parmar et al., 2021).

FID is still a strong and generally accepted indicator of the quality of generative artificial intelligence produced images. Analyzing the overlap between real and generated picture feature distributions offers a numerical assessment of visual quality and variation. Its restrictions, especially in semantic evaluation, draw attention to the requirement of complementary measures like Intersection over Union (IoU) or CLIP Score to give a more whole picture of model performance.

CLIP Score: CLIP Score is a metric that assesses the semantic alignment of textual descriptions and their related pictures. It takes advantage of OpenAI's Contrastive Language-Image Pretraining (CLIP) model, which was trained to recognize and associate images and text in a common embedding space (Radford et al., 2021). The metric is especially crucial in generative AI tasks when the main goal is to ensure that generated visuals precisely deliver the intended meaning of input prompts. Imagine you're given the task of comparing a caption to an image to see how well they match.

For example, given the text "a red apple on a wooden table" and an image depicting the same thing, you'd naturally conclude that the match is good. Assume the image depicts a green apple instead; the match is still relevant, but less precise. This is

essentially what the CLIP Score assesses: the level of semantic similarity between text and image.

CLIP consists of two neural networks—a text encoder and an image encoder. Both networks are trained to map their respective inputs (text or images) into a shared vector space. In this space, similar concepts from both modalities are positioned closer together, while dissimilar ones are farther apart. Once the text and image are converted into embeddings (numerical representations), the CLIP Score is computed by measuring the cosine similarity between these embeddings. Cosine similarity ranges from -1 to 1, where higher values indicate stronger alignment between the text and the image. For instance, the prompt "a dog playing fetch in a park" and an image of a dog chasing a ball would yield a high CLIP Score because their embeddings align closely in the shared space. Conversely, if the image showed a cat sleeping on a couch, the score would be much lower due to misalignment.

CLIP Score evaluates both textual and visual modalities simultaneously, making it uniquely suited for tasks like text-to-image generation. Unlike metrics like Fréchet Inception Distance (FID), which focus on visual quality, CLIP Score directly assesses the semantic correctness of the generated images, ensuring they match the meaning of the input prompts (Radford et al., 2021). Trained on a vast and diverse dataset, CLIP can handle a wide range of text-image pairings, making it robust for evaluating complex or uncommon prompts. Despite its strengths, CLIP Score has limitations. CLIP sometimes overemphasizes stylistic elements rather than semantic content. For example, a prompt about "a modern art painting of a sunset" might align closely with an abstract image of random shapes, as long as the colors resemble those of a sunset (Kumar et al., 2022). Since CLIP was trained on internet data, it inherits biases from the web. Prompts with culturally specific or less commonly represented concepts may yield less reliable scores (Goh et al., 2021). The effectiveness of CLIP Score is limited by the diversity and quality of its training data. Concepts not well-represented in the training set may lead to inaccurate or unpredictable results.

CLIP Score has been widely adopted in generative AI research to evaluate text-to-image models. Radford et al. (2021) demonstrated its utility in assessing the alignment between generated images and their textual descriptions, showing how it complements traditional metrics like FID. Mokady et al. (2021) applied CLIP Score to evaluate models tasked with generating images for abstract prompts, such as "a futuristic cityscape," highlighting its ability to capture complex semantic relationships. Additionally, CLIP Score has been used in fields like image retrieval, where its multi-modal embeddings enable efficient searches for images based on textual queries.

While FID measures the quality and diversity of generated images and IoU evaluates attention alignment, CLIP Score uniquely focuses on the semantic relationship between text and visuals. This makes it indispensable for applications where capturing the precise meaning of a prompt is crucial. For example, in this thesis, CLIP Score could provide complementary insights into how well Stable Diffusion represents MWEs by examining whether the generated images align semantically with idiomatic meanings. CLIP Score bridges the gap between textual prompts and visual outputs, providing a robust metric for evaluating semantic alignment. While it excels in multi-modal tasks, its sensitivity to style, cultural biases, and training data limitations underscore the importance of using it alongside other metrics like IoU and FID. Together, these metrics offer a comprehensive evaluation framework, advancing our understanding of generative models and their interpretability.

Intersection over Union (IoU): Intersection over Union (IoU) is a pivotal metric in generative AI for assessing the alignment between model-generated attention maps and predefined ground-truth regions. By measuring the overlap between predicted and actual regions of attention, IoU offers a quantitative evaluation of how effectively a model focuses on the correct areas during image generation. In this thesis, IoU plays a crucial role in evaluating the interpretability of Stable Diffusion when processing multi-word expressions (MWEs)—complex linguistic constructs that often convey meanings not directly inferable from their individual components (Miletić & Walde, 2024).

In this study, IoU is applied at the granular level to study the one-to-one correspondence of individual words in MWEs with their visual representation. This fine-grained analysis will allow us to better examine whether the model prefers to process MWEs as single semantic units or break them into two components. For instance, the idiom "spill the beans" has an IoU calculated on its components say "spill", "beans". This analysis will tell us if the model understands the idiomatic meaning of the words "spill the beans" or whether it interprets the words literally.

2.6 Ethical Considerations in Generative AI

When creating, using and implementing generative AI technologies, ethics are vital. Although these models can generate real and usable things, they create a tremendous and harmful challenge that must be dealt with to ensure their proper use. Generative AI systems have amplified concerns around bias, Cultural Appropriation And Misuse. The outputs of these systems share the nuances And ambiguities Of their training data.

One of the most serious ethical issues that generative AI faces is bias. Stable diffusion and other models are trained on datasets from across the internet and which inherently

have societal biases. These biases can appear in the outputs, promoting stereotypes or supporting bigoted beliefs. For instance, the computer might generate pictures of men with the prompt, “a doctor.” This is probably a reflection of the gender bias found in training data (Bolukbasi et al., 2016). Not just gender – Generative AI is also biased to race, age, and culture. When given culturally specific prompts, like “a wedding,” the model may simply generate a Western image instead of understanding that there are many cultural weddings in the world. The output can lead to the marginalization of underrepresented communities and world views (Binns, 2018). For multi-word expressions (MWEs), cultural misrepresentation is a serious ethical issue. Idioms often have cultural meaning that are only particular to that culture and language. The idiom, “break the ice” does not provide a direct meaning but an idiomatic one. Similarly, there may be other phrases or idioms that do not give a literal meaning either. If generative AI models misunderstand these messages, they could produce outputs that are incorrect and culturally insensitive.

When idioms are involved or there is a difference in meaning to word conjunctions, the problem of misrepresentation of cultures is aggravated. For instance, the idiom “red herring” might create an image of a literal red fish that the model will generate, if it hasn’t been trained to know that it means “distraction”. Such instances highlight the need for models that are trained on culturally diverse datasets with an ability to know when to interpret literally and when figuratively (Mishra et al., 2020).

Generative AI is in a danger of misuse because it can create realistic, often indistinguishable synthetic content. There are worries across a lot of fields about abuse or misuse of this. Generative AI can be used to create fake news and doctored images or photoshopped visuals which mislead others. For example, modifying pictures to give a false impression about an event or making deepfakes of public persons can lead to large negative effects on society (Chesney & Citron, 2019). Models inadequately trained may produce harmful, violent, or offensive content. Open access generative AI models are particularly at risk of abuses due to lack of moderation (Vincent, 2021). Generative models can produce outputs that closely resemble those of existing artworks, fuelling concerns about IP infringement and devaluation of the works of original creators (Heaven, 2020).

Addressing the ethical issues of generative AI systems requires a multi-faceted approach. Techniques like adversarial debiasing, dataset balancing and post-hoc correction can help mitigate biases. To establish trust and accountability, bias audits and transparency in dataset composition (Mehrabi et al., 2021) are critical; thereby incorporating cultural and linguistic diversification of MWEs during training can enhance the interpretation and representation of MWEs. This means using annotations

that capture cultural and contextual meanings which includes what has been done in the Multiword Expression Complexity Word Identification dataset (MWE-CWI) (Ekochmar, n.d. 2020) in this thesis. This involves establishing clear ethical guidelines and governance frameworks to mitigate misuse. It is important to have policies that help with content moderation, hold users accountable, and ensure transparency in the outputs of the model to reduce harm (Floridi et al., 2018). Ongoing studies on Diffusion Attention Attribution Maps (DAAM), Grad-CAM and LIME enhance the output of generative AI and clarify ethical issues. When researchers visualize where an image receives attention, they can see biases or misinterpretations.

Ethical safeguards must take into account the nuances of language and context for multi-word expressions. MWEs often hold hidden meanings that generative AI models must be sensitive to in order to avoid misrepresentation and harm. A phrase like “hard-pressed” or similar means difficulty or urgency. So showcasing something that may mislead those imaginations could end up offending a group of people.

This thesis aims to examine the evaluation of multi-word expressions (MWEs) by means of DAAM and IoU in Stable Diffusion, which could be useful for identifying semantic alignments and potentially ethical usages. The results highlight that generative AI systems require continuous improvement so that they could give accurate yet ethical results. The issues concerning generative AI highlight the more significant dangers involved for society. As these models are increasingly used in creative, educational, and professional spheres, it is important to deal with bias, cultural misrepresentation and misuse. If we make generative Artificial Intelligence more ethical and easy to understand, it will be a better technology in future.

The issues posed by MWEs in textual into picture generative, generative modeling, cross-attention scheme, and linguistic complication all share commonality. Studying how current models such as Stable Diffusion encode and retrieve MWE meanings provides an immediate route to unraveling the research questions set out in this thesis. In particular, we will ask (1) how DAAM can visualize and explain a model’s attention to MWEs, and (2) how a word in an MWE may shape MWE meaning. Therefore the theoretical framework developed here informs the interpretability and semantic alignment investigations that our research questions seek to answer.

3. Research Methodology

The research methodology in this thesis outlines the steps taken to solve the research problems. This chapter presents the relevant datasets and tools and describes the

computational framework that was used to analyse the interpretability of MWE in the context of stable diffusion text-to-image generation. This part of the study looks at how experts compare multiword expressions to their images, using statistical measurements including the similarity score called IoU. This chapter also delves into the iterative processes that are central to image generation, annotation, and quantitative evaluation.

This research uses a mixed-method approach. The study focuses on qualitative performing quantitative analysis on how Stable Diffusion process MWEs. The study is concerned with two issues

- 1) How can developers use Diffusion Attention Attribution Maps (DAAM) to visualize model attention for MWEs during text-to-image generation?
- 2) How do single words inside multiword expressions have semantic importance in generative models?

In order to answer these questions, the study employs computational tools, linguistic frameworks, and interpretability procedures within a systematic framework. This combination allows us to observe trends in the model's behaviour, while also providing insights that can be used to improve generative AI.

3.1 Dataset Analysis and Preprocessing

The Multiword Expression Complexity Word Identification dataset (MWE-CWI) (Ekochmar, n.d. 2020) Multicultural Expression Complexity Word Identification This dataset is useful for studying the interpretability of generative AI which include MWE processing With 4,732 annotated entries, the dataset offers a wide variety of expressions, contexts, and complexity scores which makes it ideal for the task. The building blocks of the study are basic parameters, such as Expression, Context, Annotation 'Complex probabilistic', and 'Complex binary', that is how MWEs are represented linguistically and visually.

Expression, Annotation, and Linguistic Diversity

The dataset includes the Expression field with original multi-word expressions (MWEs). These range from literal and complex phrases. Each expression is accompanied by an Annotation that categorises the expression into linguistic types (MW compounds, named entities, phrasal verbs, etc.). There are different kinds of MWEs which require a model to go against the generative nature of the model.

Figure 2 shows how The annotations are used to classify dataset types. The biggest category, Not MWE (2,180 entities), is the one we are supposed to use for simpler

expressions. The next largest group is MW Compounds (1,272 entries); e.g. mother tongue, one requiring a precise meaning. The MW Named Entities (497 entries) including Great Wall creates hurdles that require contextual and cultural knowledge. Expressions in the set include groups like verb-particle constructions, semi-fixed expressions, and prepositional phrases. The dataset is rich and giving complex example for generative modeling.

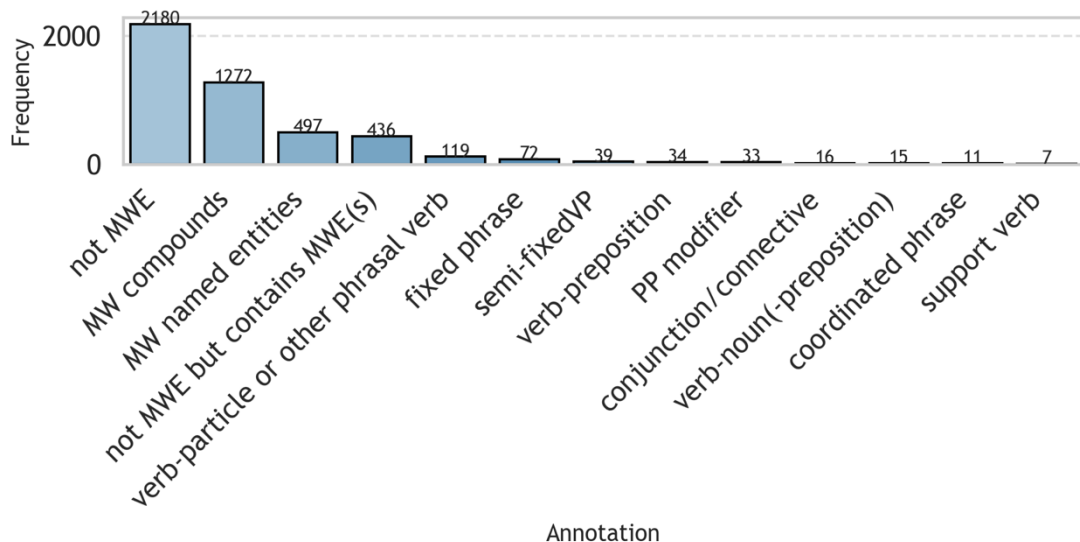


Figure 2: Frequency Distribution of Annotations in the Dataset

The importance of this linguistic diversity comes to evaluate how Stable Diffusion deals with multi-word expressions (MWEs) ranging over these various linguistic phenomena.

The Complex probabilistic score is a parameter in the dataset which indicates the number of annotators indicating the expression as complex. The estimated difficulty of the Multiword Expressions (MWEs) can be quantified using a scale, which is expressed as a probability that ranges from 0.0 to 0.8. As shown in Figure 3, the organisation of scores marks that there is quite an interesting trend and reveals how much of a challenge MWEs are for generative models.

The histogram (Figure 3) shows fluctuating trends of complexity score assigned to multi-word expressions (MWEs). Most expressions were very easy for the annotators, as indicated by a low complexity score (0.0-0.2). A literal phrase such as ‘morning coffee’ has a logical meaning which is easy to comprehend. On the other hand, the MWEs with high complexity scores (0.4-0.6) are idiomatic or culturally specific. For example, the phrase “domestic violence” (complexity score: 0.6) will take some outside knowledge for the receiver to fully grasp. Generative models must do more

than take things literally; they must also express themselves in a figurative or idiomatic manner.

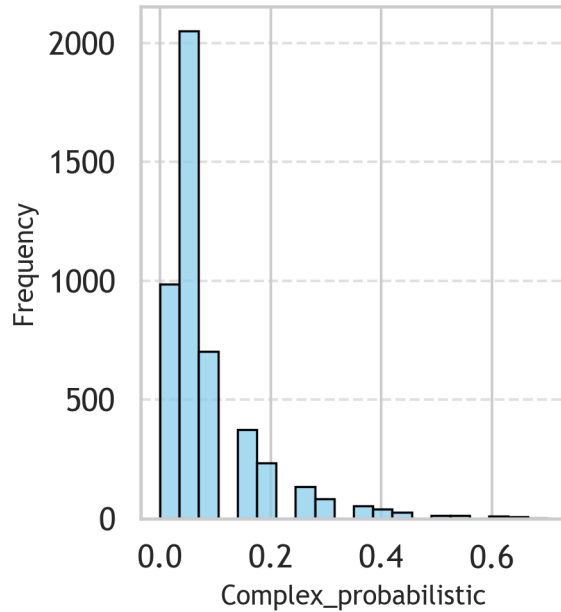


Figure 3: Distribution of Complex Probabilistic Values in the Dataset

The Complex binary parameter adds to the analyses by giving a binary classification of complexity; 1 means complex, while 0 means simple. This two-value metric enables MWE grouping into two categories to allow for analyzability in a formal fashion.

3.2 Context and Prompt Design

Acting as a link between textual inputs and visual outputs, prompt engineering is fundamental in text-to-Image generation. Working with multi-word expressions (MWEs), which may have idiomatic or context-specific meanings that can test generative models like Stable Diffusion, this becomes especially crucial. This work adapted the contexts from the Multiword Expression Complexity Word Identification dataset (MWE-CWI) (Ekochmar, n.d. 2020) into succinct but significant prompts using prompt engineering. These stimuli were made to stay inside Stable Diffusion's 77-token input limit while preserving MWEs' fundamental meaning. This method made it possible to evaluate the model's capacity to match textual descriptions with matching images with a targeted eye toward. Research on generative artificial intelligence has revealed that how models understand and create images can be much influenced by the quality of stimuli. Studies by Li et al. (2023) and Alaluf et al. (2022) emphasize this fact. Li and colleagues underline that well-crafted questions serve as anchors, guiding the

model toward important textual parts and generating coherent images. Likewise, especially for culturally complex or idiomatic terms, Alaluf and co-authors discovered that exact and explicit cues enhance the interpretability and correctness of produced outputs. These results guide the approaches applied in this work by proving the need of efficient prompt engineering for MWEs.

Stable Diffusion gains much from well-designed cues since it depends on cross-attention techniques to link textual inputs with visual elements. In this work, carefully crafted prompts derived from Multiword Expression Complexity Word Identification dataset (MWE-CWI) (Ekochmar, n.d. 2020) annotated environments. These settings were streamlined to keep their significance while following the token restriction. The phrase “domestic disturbance,” for instance, first surfaced in this context: "Goodyear said police previously responded to the Florida 'Space Coast' home, about 15 miles south of Cape Canaveral, for domestic disturbance calls involving Jaxs Johnson." This was changed down to the prompt: "A Florida 'Space Coast' home experiencing domestic disturbances requiring police attention." This version makes the input more fit for the model while maintaining the important facts and significance. (OpenAI, 2023)

“Presidential nomination” serves as another example that required balancing its literal and abstract meanings. Initially framed as: “Martin, best known for spreading multiple conspiracy theories, announced his presidential nomination,” it was adapted to: “A candidate announcing their presidential nomination during a press conference.” This approach ensured that the essence of the phrase remained intact while adhering to token constraints. (OpenAI, 2023)

These examples highlight the delicate balance required in effective prompt engineering, where responses must be both precise and concise. Idiomatic MWEs demand particular focus to preserve their metaphorical meanings while maintaining essential context. This study tackles interpretability challenges, enhancing Stable Diffusion's ability to produce accurate visual representations by systematically refining these complex phrases into optimal prompts. This method not only reveals areas for further improvement but also demonstrates the model's capacity to handle linguistic nuances effectively.

3.3 Stable Diffusion Setup and Implementation

Stable Diffusion 2.1 is a novel application of latent diffusion models, tailored for producing high-resolution images from textual descriptions. Its capacity for semantically coherent images makes it a useful framework for exploring the interpretability of multi-word expressions (MWEs). Variational Autoencoders (VAEs),

denoising diffusion probabilistic processes (DDPMs), and cross-attention mechanisms are combined in the model as described in Subsection 2.2 and shown in Figure 1. Together, these elements let textual inputs be mapped to visually relevant outputs (Rombach et al., 2022). Emphasizing their importance in achieving the goals of the thesis, this part describes the pragmatic implementation techniques used in this work.

Using open-source tools including Hugging Face Transformers for text processing and PyTorch for neural network modeling, the implementation took place in a Python environment. Using NVIDIA Tesla T4 GPUs to fulfill the computational needs of high-resolution image production and the analysis of attention maps with Diffusion Attention Attribution Maps (DAAM), the tests were conducted on Google Colab Pro.

Using Stable Diffusion's text encoder, key implementation phases consisted in tokenizing and embedding textual inputs from the Multiword Expression Complexity Word Identification dataset (MWE-CWI) (Ekochmar, n.d. 2020). This stage guaranteed that the semantic subtleties of MWEs were maintained, therefore enabling the model to precisely match idioms and context-dependent meanings with matching visual outputs using its cross-attention systems.

Dealing with the 77-token input limit of Stable Diffusion presented one of the main technical difficulties. MWEs usually arise in sophisticated or verbose settings outside this restriction. The original contexts were methodically reduced into short cues while maintaining their semantic integrity to guarantee fit.

Integration of Diffusion Attention Attribution Maps (DAAM) into the implementation helped to investigate how Stable Diffusion processes MWEs. Visualizing the cross-attention distributions of the model, DAAM overlays attention weights onto produced visuals. These heatmaps show visual areas connected to particular tokens or phrases, therefore offering a detailed picture of the interpretive process of the model (Tang et al., 2022).

When examining the term "hard pressed," DAAM visualizations, for example, showed whether the model's attention was concentrated on the metaphorical meaning—that which denoted difficulty—or scattered among its numerous constituents. Such insights were essential for assessing the alignment between textual inputs and visual outputs, therefore supporting the more general interpretability goals of the thesis.

Large-scale testing and analysis were greatly aided by Google Colab Pro, with its NVIDIA Tesla T4 GPUs. The GPUs let high-resolution picture creation, effective textural embedding processing, and DAAM heatmap computing possible. Supported iterative improvements in rapid design and interpretability analysis, the cloud-based architecture guarantees a methodical and scalable approach to answer the research

objectives. The interpretability problems presented by MWEs are addressed in this thesis by means of a customized configuration and Stable Diffusion implementation. This study evaluates Stable Diffusion's capacity to comprehensively interpret MWEs by transforming complex linguistic constructs into optimal prompts and employing DAAM for in-depth attention analysis. These approaches not only fit the thesis goals but also help to clarify how generative artificial intelligence systems control linguistic complexity, therefore opening the path for developments in semantic and visual coherence in generative modeling.

3.4 DAAM Integration for Interpretability

This thesis investigated Stable Diffusion's handling of multi-word expressions (MWEs) using Diffusion Attention Attribution Maps (DAAM) as a fundamental interpretability tool. By creating heatmaps that overlay attention weights onto created images, DAAM offers a thorough visualizing of the cross-attention distributions of the model. These heatmaps provide vital information on the areas of a picture that match tokens in the input text, therefore allowing a precise knowledge of how the model aligns textual and visual representations (Tang et al., 2022).

The DAAM technique consists in various stages that taken together generate important visualizations. Cross-attention weights are extracted from many layers and timesteps during latent diffusion model denoising. These weights show how much particular textual tokens affect various areas of the produced image. This extraction emphasizes for MWEs how the model allocates its attention among individual words and their collective meaning. Cross-attention happens across several layers of Stable Diffusion, each adding to the final image at different timesteps. DAAM compiles these weights to produce a single focus of the model. This aggregation guarantees that the model's wider decision-making process is considered rather than only one layer, therefore ensuring that neither is constrained. Heatmaps displaying token-specific attention distributions are created by projecting the aggregated attention weights onto the spatial dimensions of the produced image. These maps let academics exactly identify which areas of an image match particular words or phrases found in the input text.

Because of its non-compositional character and context-dependent interpretations, MWEs provide special difficulties for generative models. By separating between two forms of attention, DAAM offers a detailed perspective on how Stable Diffusion analyzes these expressions.

DAAM heatmaps show focused attention on aspects that reflect the idiomatic or contextual meaning of the utterance in circumstances when the model appropriately

interprets a MWE as a single semantic unit. For instance, the term "domestic disturbance" reflects the idiomatic sense of the phrase by ideally producing a heatmap emphasizing on visual features such a house with police activity. (Figure 4 (a))

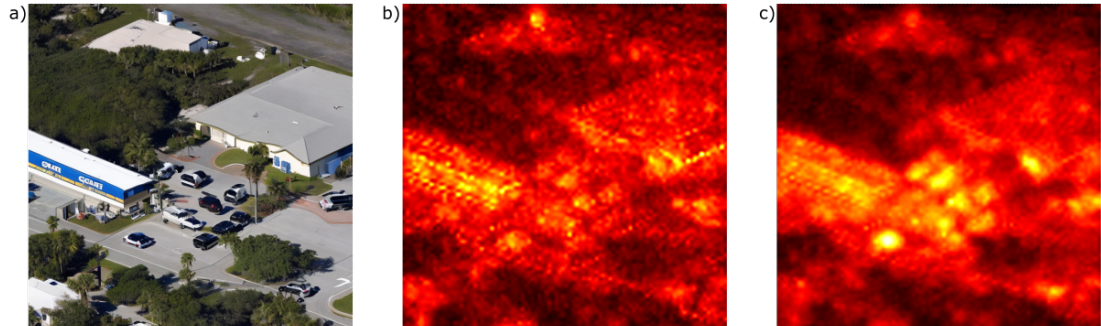


Figure 4: Visualization of the Stable Diffusion model's interpretation of the phrase "domestic disturbance": a) The generated image b) DAAM heatmap for domestic; c) DAAM heatmap for disturbance

Ensuring that the produced image corresponds with the intended meaning of MWEs—especially those with abstract or culturally particular interpretations—depends on holistic attention. (Figure 4 (b), (c))

Component-Level Attention: DAAM also highlights scenarios where the model focuses on individual words within a multi-word expression (MWE) rather than capturing their combined or holistic meaning. For instance, in the prompt "A man riding a cycle," if the model isolates its attention to "man" and "cycle" independently, the resulting heatmaps show separate overlays—one highlighting the "man" and another focusing on the "cycle." (figure 5 (a), (b), (c)) This fragmentation indicates that the model interprets the words as distinct entities rather than understanding their interconnected meaning, potentially leading to disjointed or incomplete visual representations.

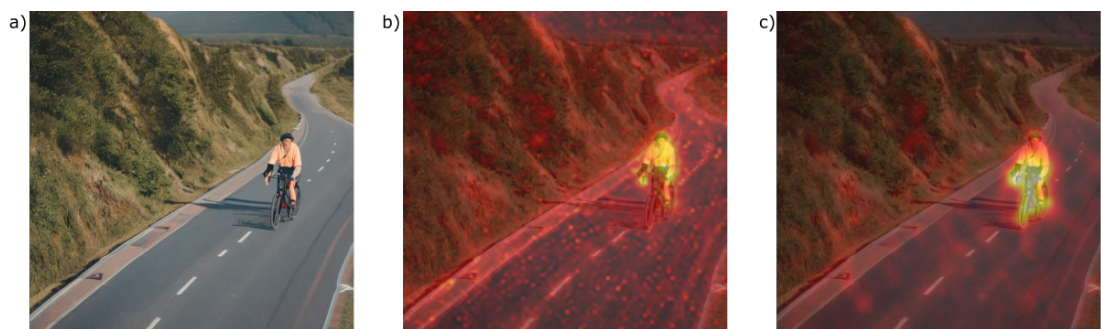


Figure 5: Visualization of component-level attention for the prompt A man riding a cycle: a) Generated image by the Stable Diffusion model b) DAAM attention map for the word “man” c) DAAM attention map for the word “cycle”

Although DAAM offers insightful analysis, use of it presents several difficulties. Ensuring proper aggregating of attention weights across layers is one obvious problem. Variations in attention distributions at various timesteps might produce heatmaps either underrepresenting or overstretching particular textual tokens. Dynamic normalizing methods were used to solve this such that the resulting heatmaps show balanced attention distributions.

Interpreting DAAM heatmaps for unclear MWEs presents still another difficulty. Expressions like “take credit,” for instance, might have several reasonable visual interpretations based on the context. DAAM visualizations for such words demand thorough study to ascertain if the model's attention follows the intended meaning or veers into literal interpretation. In this thesis, DAAM proved to be essential for assessing Stable Diffusion interpretability in processing MWEs from the Multiword Expression Complexity Word Identification dataset (MWE-CWI) (Ekochmar, n.d. 2020) . For instance, DAAM examined the word “hard pressed” to see whether the model handled it literally as two separate ideas or caught the idioms of difficulty and pressure. The created heatmaps showed whether the generated attention was focused on pertinent visual elements, such a person in a stressful circumstance, or scattered over unrelated areas.

The inclusion of DAAM into this work corresponds with past work using attention attribution strategies to improve generative model interpretability. Studies on the value of attention-based visualizations in comprehending model behavior by Chefer et al. (2021) and Selvaraju et al. (2017) highlight These analyses show that methods such as DAAM not only offer understanding of model performance but also help to improve model architectures by pointing up areas of attention mechanism enhancement needs.

The results of DAAM studies for this thesis transcend the setting of MWEs. Especially for linguistically complicated inputs, they draw attention to the more general difficulties generative artificial intelligence systems have matching textual and visual modalities. The knowledge acquired from DAAM could guide next developments in the cross-attention mechanisms of Stable Diffusion, so permitting more accurate representations of idiomatic and context-dependent language. DAAM closes the distance between textual cues and their visual representations by presenting a clear picture of the interpretative process of the model, therefore laying a basis for more interpretable and dependable generative artificial intelligence systems. This thesis

uses DAAM not only as a means of improving the knowledge of how MWEs are handled in state-of-the-art modern text-to-image models but also as an analytical tool.

3.5 Evaluation Metrics

This thesis uses Intersection over Union (IoU) as the main metric to assess the interpretability and semantic correctness of Stable Diffusion in processing multi-word expressions (MWEs). Evaluating the alignment between the attention of the model and the intended semantic meaning of MWEs depends on a consistent quantitative measure, IoU. Using DAAM-generated heatmaps and binary masks, IoU provides precise understanding of the degree to which Stable Diffusion understands MWEs holistically or as a collection of discrete components. This work also places its results within the larger framework of generative artificial intelligence evaluation by using knowledge from past work on Fréchet Inception Distance (FID) and CLIP Score. Although this thesis did not specifically apply these measures, they offer insightful comparisons on the difficulties and achievements of MWE representation in generative modeling.

The overlap between visual regions associated with different words in MWEs and the attention heatmaps generated by DAAM is quantified using Intersection over Union (IoU). In this study, heatmaps for individual words within MWEs were dynamically normalized to highlight regions of highest attention. These heatmaps were then thresholded to create binary masks, effectively identifying areas where the model focused its attention. IoU was calculated as the ratio of the overlapping area (intersection) to the total area covered by attention (union) between the binary masks. The IoU analysis for MWEs was conducted in two stages. The Holistic IoU evaluated how well the model's attention distribution for the entire MWE aligned with its corresponding visual representation. High IoU scores indicated that the model successfully interpreted the idiomatic or context-specific meaning of the MWE, rendering it as a unified concept. Conversely, the low IoU analyzed the attention distributions for individual words within the MWE.

In the case of the MWE travel document, provided in the context, “There was also a Syrian travel document found, police spokesman Hans Peter Doskozil told journalists, so, of course, our first assumption is that these people were migrants, likely a group of Syrian migrants,” DAAM generated two separate heatmaps: one focusing on “travel” (Figure 4(b)) and the other on “document” (Figure 4(c)). This analysis highlights how the model distributes attention across individual components of the MWE within a given context.

To calculate the IoU, these heatmaps were transformed into binary masks and matched. A high Holistic IoU score indicates that the model understands the term as a coherent semantic entity, hence probably producing images like a passport or another travel-related document. Conversely, low Holistic IoU with high Component-Level IoU showed that the model treated "travel" and "document" independently, likely producing disconnected pictures of a suitcase and a sheet of paper, failing to capture the idioms essence.

To focus areas of great interest, these heatmaps are dynamically thresholded at a chosen percentile, 90 percentiles, after normalizing. While all others are set to 0 (black) (Figure 4(f)), pixels over this threshold are set to 1 (white) (Figure 4(d)), hence producing binary masks highlighting the areas of intense model attention for each word.

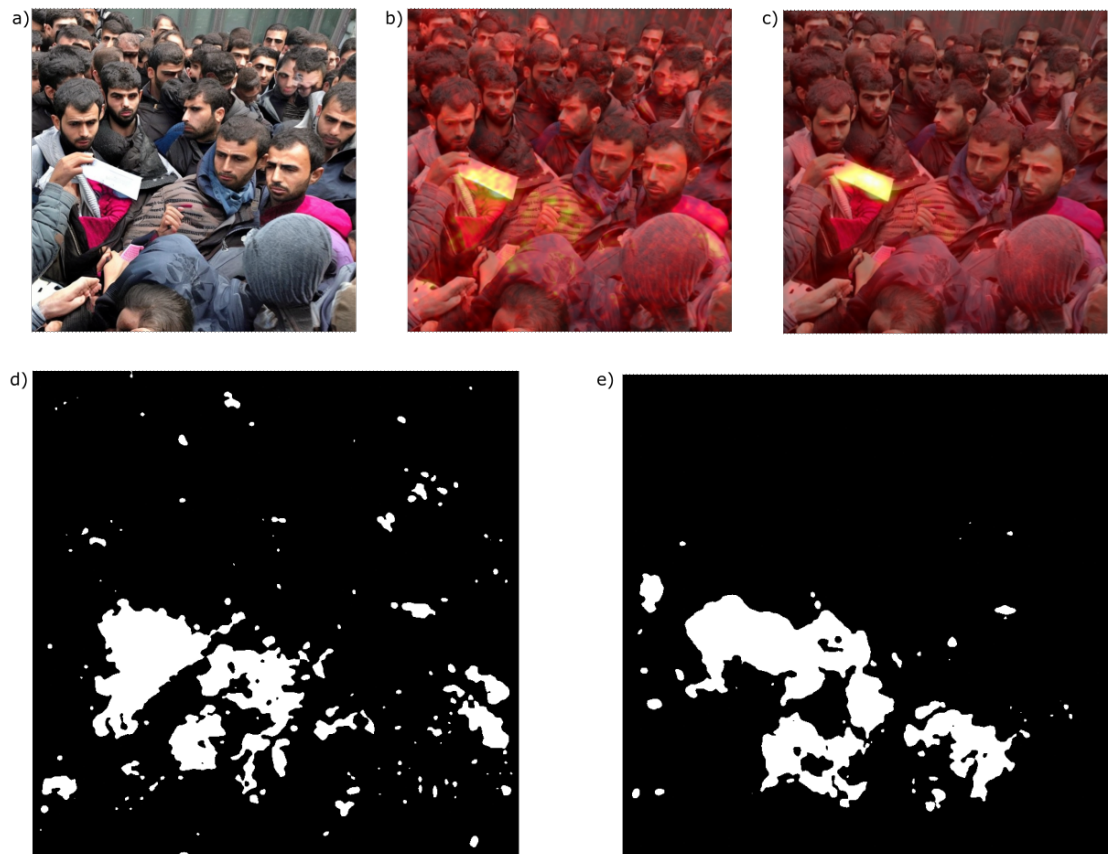


Figure 6: Visualization of DAAM Attention and Binary Mask Creation for "Travel Document"

- (a) The image generated by the Stable Diffusion model for the MWE
- (b) DAAM attention heatmap showing the model's focus on "travel."
- (c) DAAM attention heatmap highlighting the model's focus on "document."
- (d) Binary mask derived from the "travel" heatmap after thresholding.
- (e) Binary mask derived from the "document" heatmap after thresholding.

IoU	Complex_probabilistic	Expression	Context	Annotation
0.3885	0.05	travel document	"There was also a Syrian travel document found" police spokesman Hans Peter Doskozil told journalists "So of course our first assumption is that these people were migrants, and likely a group of Syrian migrants."	MW compounds

Table 1: Example Annotation from the Multiword Expression Complexity Word Identification dataset (MWE-CWI) (Ekochmar. n.d. 2020)

Once the binary masks are created, the IoU is calculated by comparing every pair of words in the MWE. By counting overlapping white pixels in the binary masks, we determine the intersection.

The value of IoU will be zero if there is no union (attention) at all. The IoU values of each pair of words are saved in a dictionary where the keys are word pairs, and the values are their respective IoU values. The text file and the excel file is saved in the Directory for keeping record.

An example shows this process. Consider two binary masks.

$$\begin{array}{ccc}
 & 1 & 1 & 0 \\
 \text{Mask 1:} & 1 & 0 & 0 \\
 & 0 & 0 & 0
 \end{array}
 \qquad
 \begin{array}{ccc}
 & 1 & 0 & 0 \\
 \text{Mask 2:} & 1 & 1 & 0 \\
 & 0 & 0 & 0
 \end{array}$$

The region of overlap contains pixels with a value of 1 in either mask. It consists of 2 pixels. The union includes 4 pixels that are white in either mask. The IoU is calculated as:

$$\text{IoU} = \frac{\text{Intersection}}{\text{Union}} = \frac{2}{4} = 0.5$$

Thus, we carry on with the whole process with every word pair in the post made dataset for granular evaluation of how accurately the model’s attention reflects the true semantic meanings of MWEs.

Talking of the IoU, Stable Diffusion is seen to be quite proficient in interpreting literal MWEs-like “morning coffee”, which got high Holistic IoU scores throughout. However, MWEs that are idiomatic or culturally restricted, like “same-sex marriage”, often

showed fragmented attention distributions, leading to lower Holistic IoU scores. The study showed that Stable Diffusion prefers literal descriptions rather than idioms or abstract meanings which is something that was found in other generative AI studies too.

The insights developed from IoU besides FID and CLIP Score in similar research broadened the comprehension of the MWE challenges. FID is a statistic used to tell how real your generated images are. It looks at the feature distribution to see how similar they are. The study by Heusel et al. (2017) and Mokady et al. (2021) demonstrate that MWEs whose meanings are simple and literal, such as ‘blue butterfly’, have a low FID score (around 12.5), indicating generative models visualize realistic results for them. On the other hand, idiomatic MWEs like spill the beans had higher FID scores (~20.3), meaning they were of lower quality visually as the models failed to illustrate something abstract or idiomatic. Like the IoU results in this thesis, fragmented attention distributions were often observed for fixed phrase MWEs.

In the same way, the CLIP Score gave complementary insights. By looking at the cosine similarity of the text and image embeddings in a single latent space to measure semantic alignment. Research by Radford et al. and Gal et al. demonstrated that Literal MWEs like red apple were guaranteed to achieve high CLIP scores (close to 0.85) as these phrases were semantically closely aligned with the images they refer to. On the contrary, fixed phrases like “sense of shock”, scored 0.2387. This misalignment showed that the generated images reflected the literal meaning of the expression rather than the idiomatic meaning. The findings further validate the earlier IoU analysis conducted in this thesis and emphasize the limitations of Stable Diffusion in effectively interpreting idiomatic and contextually dependent MWEs.

Stable Diffusion does well with MWEs that have their literal meaning. However, it fails to produce coherently represented MWEs that are idiomatic or abstract in meaning. According to an analysis of IoU, FID, and CLIP Score of generative AI systems, the AI is still not able to understand and generate complex language images. The findings in this thesis are consistent with past studies, suggesting a need for improvements in training data and model architectures. Improving the model’s semantic interpretation of MWEs could make it better at connecting text and images, if such advancements come about. Changes in the model would reduce biases and the complexity of language models. This will help the model get a better understanding of human preferences.

In conclusion, the methodological framework presented in this study establishes a structured approach for analyzing how Stable Diffusion interprets MWEs. The process begins with the extraction of a MW components set of MWEs, for which corresponding

images are generated. DAAM is then applied to these images to produce attention heatmaps, revealing the areas of the image most influenced by the individual tokens within the textual prompt (Figure 4(c), (d)). These heatmaps are subsequently thresholded to create binary masks that capture regions of highest attention intensity.

This methodology transitions from the conceptual formulation of research questions to their operationalization by mapping these binary masks onto pre-defined regions corresponding to the intended meaning of the MWEs and calculating Intersection over Union (IoU) scores (Table 1, Table 4). This approach enables precise evaluation of how closely the model's attention distributions align with the intended semantic meanings of MWEs and identifies the contribution of individual words to the overall interpretation. By integrating DAAM and IoU into a coherent analytical pipeline, this framework bridges the theoretical underpinnings outlined earlier with the results and analysis discussed in subsequent sections, providing a robust mechanism for evaluating the interpretability and semantic fidelity of generative AI models.

4. Results

This chapter presents the outcomes derived from the methodological framework, thereby directly linking the results to the study questions. The results are organized based on a validation of the methodological approach to increasingly complex MWE evaluations. This method ensures the relevance of the selected techniques DAAM and IoU and offers insight into Stable Diffusion's ability to interpret and portray MWEs with varying levels of linguistic complexity. The results are systematically presented to offer a comprehensive overview. A straightforward observation of "a blue Audi on the German street" facilitates the validation of the strategy. This example demonstrates the efficacy of DAAM in calculating paired IoU scores for the terms "blue," "Audi," and "vehicle," while also illustrating attention distributions. This functions as a proof of concept, illustrating that the structure operates as intended and establishing a foundation for more comprehensive research. Subsequently, we will analyze the performance of 500 MW components from the dataset through visual representation and scatter plots. We'll look at how this framework works with non-MWEs, finally evaluating it across all categories.

4.1 Demonstrating the Methodology with a Simple Example

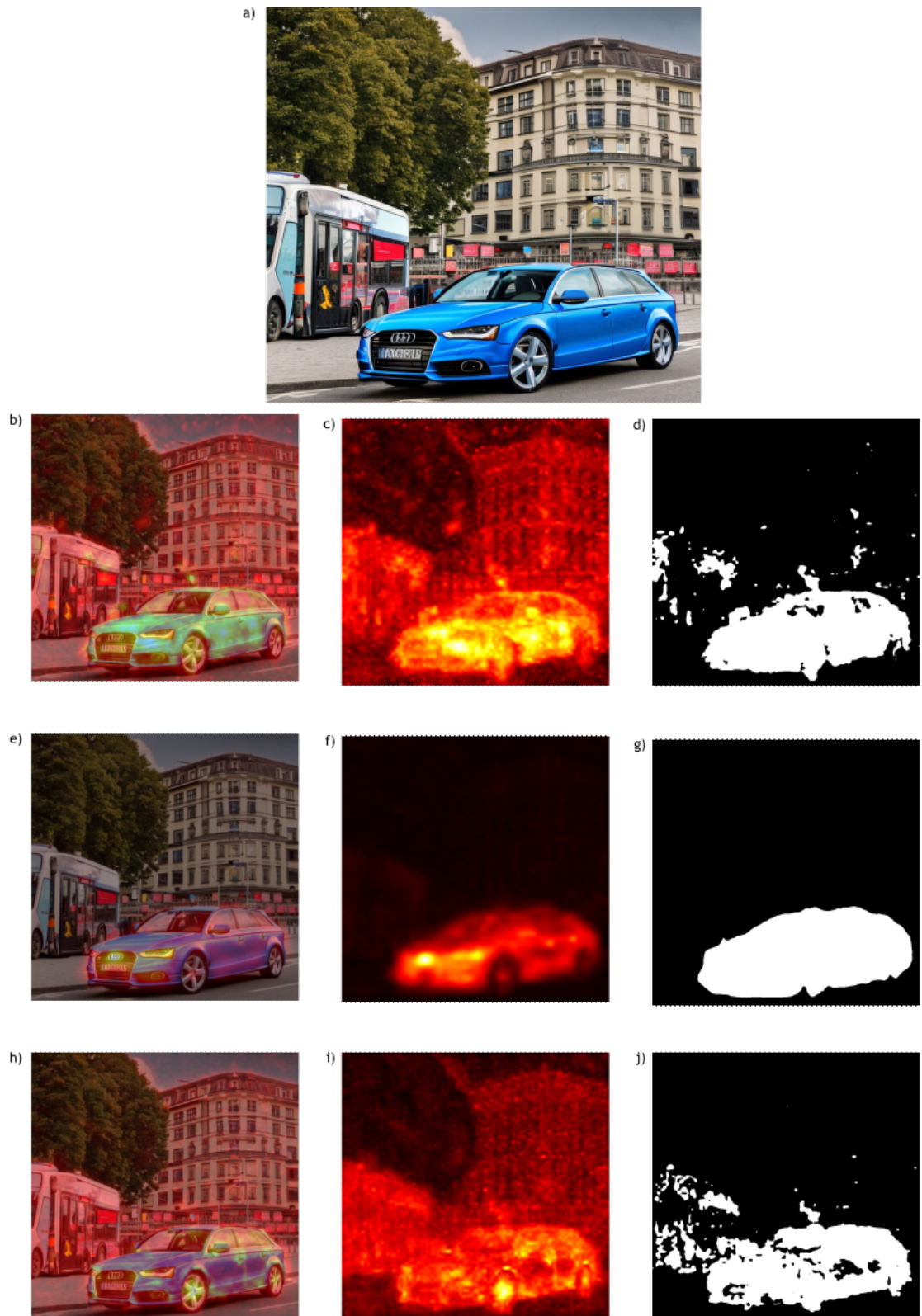


Figure 7: DAAM Analysis for A Blue Audi Vehicle on the German street a) Generated image by model b)DAAM overlay for blue c) Heatmap for blue d) binary mask for blue, e)DAAM overlay for Audi f) Heatmap for Audi d) binary mask for Audi, b)DAAM overlay for vehicle c) Heatmap for vehicle d) binary mask for vehicle

A straightforward example was utilized to validate the methods employed in this thesis: “a blue Audi vehicle on the German street.” This example illustrates the integration of Stable Diffusion, DAAM (Diffusion Attention Attribution Maps), and IoU computations, functioning as a proof of concept for the research pipeline.

The produced image (Figure 7a) depicts a blue Audi traveling on a German street, consistent with the prompt's description. Heatmaps generated by DAAM illustrate the distribution of attention across specific terms: “blue” (Figure 7 b), “Audi” (Figure 7 e), and “vehicle” (Figure 7 h). The heatmaps illustrate the model's emphasis on critical visual attributes such as the vehicle's color, brand logo, and shape. Table 2 displays the pairwise Intersection over Union (IoU) results.

Multiword Pair	Intersection over Union
Blue, Audi	0.7171
Blue, vehicle	0.6700
Audi, Vehicle	0.6677

Table 2: Pairwise IoU value for A Blue Audi Vehicle on German street

The results confirm that the pipeline successfully links text to corresponding images. This example lays the groundwork for understanding more complex MWEs.

4.2 MW Compounds Interpretation

Using 1,272 annotated entries from the dataset, Stable Diffusion's ability to manage Multi-Word Expressions (MWEs) in the category of MW compounds was evaluated. From this subgroup, 500 MWEs were randomly selected for thorough investigation with an eye toward their generated images and related IoU values to evaluate the model's performance in capturing their general meanings. I will demonstrate how the following MWEs “thermonuclear fusion,” and “crime scene investigation” shown the model's ability to precisely assess MWEs and give coherent visual representations for improved understanding.

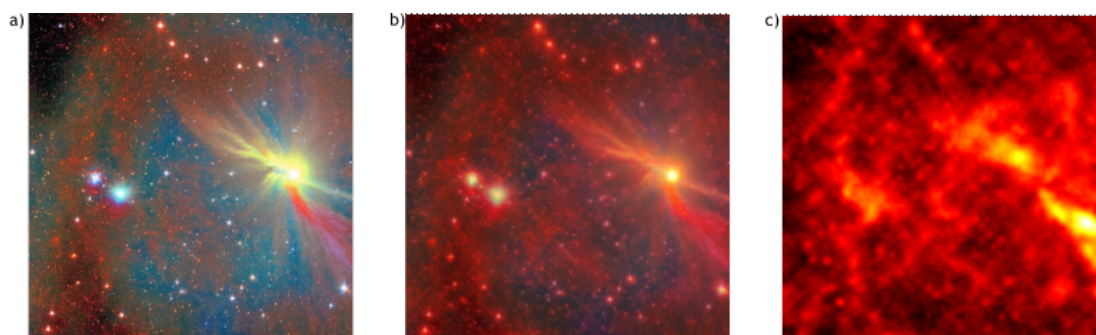


Figure 8: Visualization of the Stable Diffusion model's interpretation of the phrase thermonuclear fusion: a) The generated image, b) DAAM attention map for thermonuclear, c) DAAM attention map for fusion.

Derived from the Multiword Expression Complexity Word Identification dataset (MWE-CWI) (Ekochmar, n.d. 2020), the mwe "thermonuclear fusion" with the contextual description "High-quality spectral data acquired by the Keck 1 telescope in November 1995 showed that Teide 1 had kept the initial lithium amount of the original molecular cloud from which Pleiades stars formed, proving the lack of thermonuclear fusion in her core." This MWE was expected to produce a consistent visual depiction of the scientific and astrophysical relevance of "thermonuclear fusion" by the Stable Diffusion model. Consistent with its colloquial and technical meaning, the model should show an image of a star-like object or plasma environment indicating fusion activities. The model produced a vivid astrophysical landscape in Figure 8a, including a visually arresting nebular cloud and what looked to be strong energy bursts. Though it lacks specific scientific or visual clues, such explicit fusion reactions or star core characteristics, this picture somewhat corresponds with the conceptual core of thermonuclear fusion. Presumably in line with the energetic connections of the word, the overlay (Figure 8b) displayed focused attention on bright and lively portions of the nebula. With an eye toward similar areas—especially the brilliant core area of the nebular representation—the attention map for "fusion" (Figure 8c) suggested the model's attempt to link the phrase with ideas of energy convergence. The IoU for the pair ("thermonuclear," "fusion") came out to be 0.5206. The attention maps and the spatial areas of interest in the produced image show a fair degree of alignment based on this modest score. Although the produced output and DAAM representations fit thematically with "thermonuclear fusion," the IoU score shows shortcomings in the model's semantically binding power.

Let us now delve further into another instance whereby three words exist in mwe. Here we computed the pairwise IoU computation. Derived from the Multiword Expression Complexity Word Identification dataset (MWE-CWI) (Ekochmar, n.d. 2020),

the Multi-Word Expression Crime scene investigators, accompanied by the contextual description, “A spokeswoman for Merseyside Police said: ‘Crime scene investigators are in attendance at the scene, and officers are appealing to anyone who may have seen anyone acting suspiciously on Bidston Hill over the last few days to come forward,’” was evaluated for its representation within Stable Diffusion. Usually involving investigators at a crime scene, sometimes with related forensic or police activity, the phrase is intended to produce a visual representation that captures the investigation core.



Figure 9: Visualization of the Stable Diffusion model's interpretation of the phrase Crime scene investigators: a) The generated image, b) DAAM attention map for Crime, c) DAAM attention map for scene, d) DAAM attention map for investigators

Capturing law enforcement officials at what looks to be a secured crime scene, the produced image (Figure 9a) depicts a realistic depiction of a police activity zone. This shows the model's capacity to capture the contextual core of the MWE, therefore complementing the expected visual semantics of the prompt.

Further understanding of the attention distribution for certain phrases inside the MWE comes from the DAAM overlays: Crime (Figure 9b): Indicating an attempt to link the phrase with the law enforcement environment, the overlay for crime concentrates mostly on the scene's investigating region, especially near the police car and the officers. Consistent with the geographical context of a crime scene in real-world events, attention for scene encompasses the larger region outlined by the crime tape (Figure 9c). The overlay for investigators mostly draws attention to the law enforcement personnel shown in the picture, therefore illustrating the connection of the term with persons in investigative capacity (Figure 9d).

Multiword Pair	Intersection over Union
Crime and scene	0.5596
Crime and investigators	0.6697
Scene and investigators	0.5440

Table 3: pairwise Intersection over Union for Crime scene investigators

These IoU values show in Table 3 rather good correspondence between the textual elements and their visual equivalents. Especially, the high IoU for Crime and investigators implies that the model effectively correlates the crime-related activities with the presence of investigators. The somewhat lower IoU for scene and investigators. However, indicates that although the geographical distribution catches important features, some subtleties of the MWE's overall meaning may not be totally incorporated. All things considered, the produced image and its matching DAAM overlays fit rather nicely the desired thematic and contextual portrayal of Crime scene investigators. The IoU values show a good degree of semantic binding, therefore highlighting the capacity of the model to comprehend MWEs with investigative and procedural settings.

4.3 Quantitative Insights into IoU Performance and Complexity Dynamics in MWE Interpretation

Especially in the context of multi-word expressions (MWEs), the study of IoU values across several annotation categories offers insightful information on the performance of the model in matching textual prompts with visual outputs. Examining the link between IoU and Complex probabilistic on a scatter plot (Figure 10) shows a weak connection devoid of any obvious linear or nonlinear trend. While Complex

probabilistic scores remain mostly below 0.5, implying that the probabilistic complexity of MWEs does not greatly influence the capacity of the model to produce accurate picture alignments, IoU values mostly cluster between 0.2 and 0.6. This lack of a clear association suggests that elements outside of complexity, such as complex binary, contextual subtleties or annotation traits, most certainly influence IoU performance more significantly. Especially, scatter plot outliers with either high IoU or high complexity draw attention to isolated events where the model either performed very well or failed to visually align complicated sentences. These events call for qualitative investigation to spot possible deviations in model behavior.

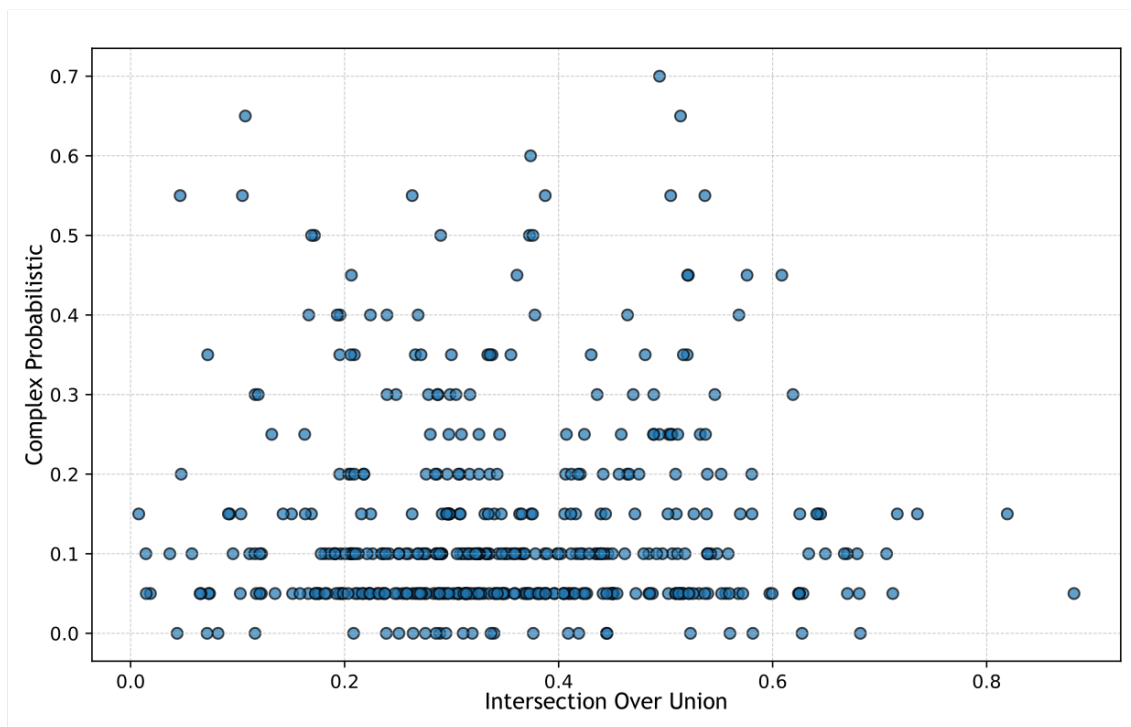


Figure 10: Scatter Plot of IoU vs. Complex Probabilistic Scores for Multi-Word Expressions

The box plot (Figure 11) shows the distribution of IoU values over annotation categories, so augmenting this study. Among these, MW compounds show the most variation; IoU values range from 0.01 to 0.67, so reflecting the several linguistic and semantic difficulties within this group. This variety implies that MW compounds comprise idioms, literal, and ambiguous language, each of which presents different degrees of interpretability for the model. Conversely, MW named entities and not MWE categories show tighter distributions with medians somewhat above 0.3, suggesting that simpler structures—such as named entities—are more regularly aligned by the model. Reflecting the difficulty of reading and aligning MWEs buried inside more general non-MWE settings, the category not MWE but contains MWE(s) has the lowest median IoU (~0.29) and a limited range.

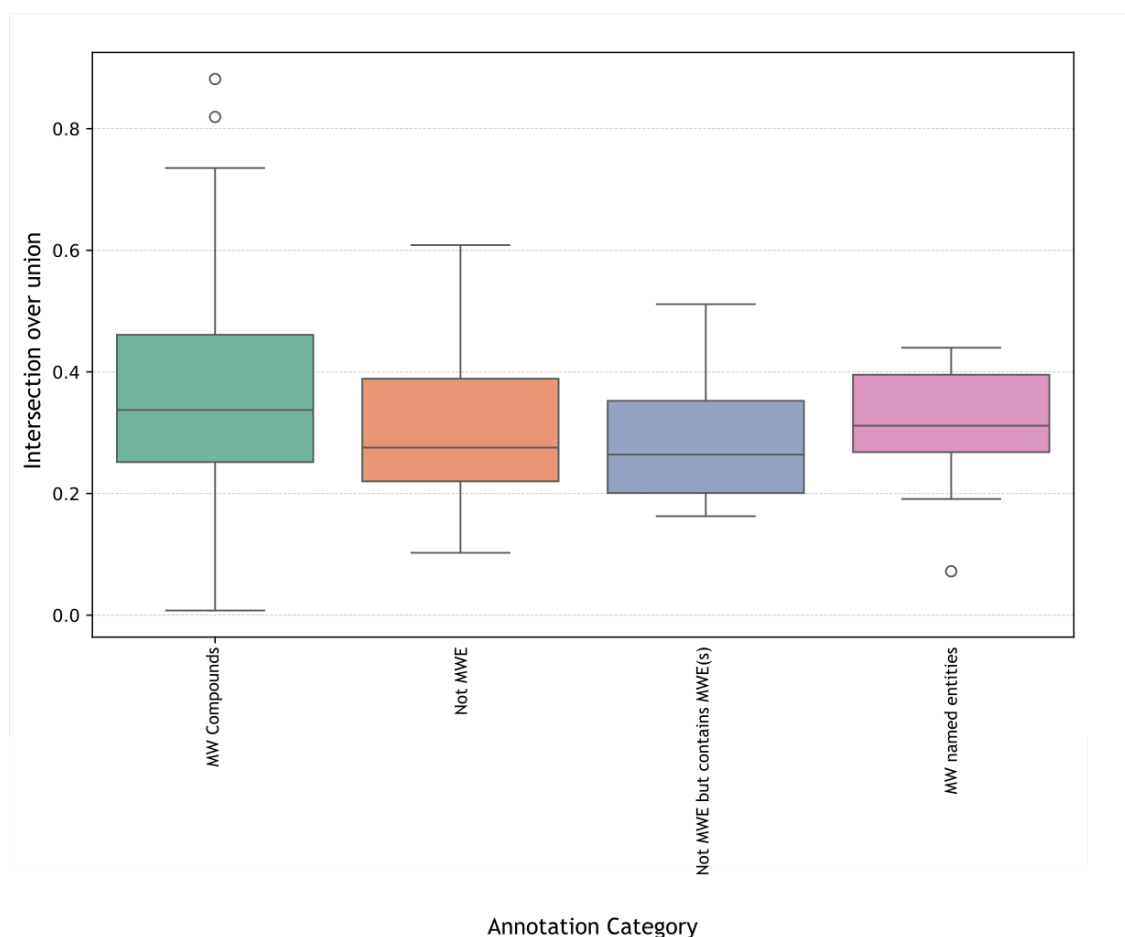


Figure 11: Box Plot of IoU Distribution Across Annotation Categories

Annotation	Expression	Complex probabilistic	Intersection Over Union
MW compounds	life threatening	0.55	0.3874
MW named entities	Economic Zone	0.2	0.4181
not MWE	fatally injured	0.45	0.6085
MW compounds	Dangerous driving	0.05	0.4876
MW compounds	domestic disturbance	0.45	0.5761
MW named entities	Formica fusca	0.35	0.0720
MW compounds	brown dwarf	0.15	0.5698
MW compounds	transport minister	0.05	0.316
MW compounds	same-sex marriage	0.05	0.1988

MW compounds	aqua vitae	0.5	0.1718
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Table 4: Comparison of Entities with High and Low IoU Scores Based on Annotation and Complexity

Particular instances support these conclusions. Expressions such domestic disturbance, Dangerous driving, crime scene investigators, thermonuclear fusion, brown dwarf clearly show the success of the model in capturing the semantic and visual essence of MWEs. Conversely, presidential nomination, transport minister, Blackpool bid, Laboratory mice, and financial aid reflect circumstances where the model struggled to generate coherent visual representations, particularly for abstract or context-specific phrases. (Table 4)

These results underline the need of improving attention mechanisms to manage the variety of MWEs more efficiently and of augmenting datasets to better reflect difficult situations like embedded MWEs. Moreover, the great diversity in MW emphasizes the need of focused error analysis to find certain situations in which the model performs either satisfactorially or badly, hence guiding future improvements. Although the model shows some degree of alignment capacity, overall the results imply that the complete semantic complexity of MWEs is still difficult to capture, thereby stressing the need of focused enhancements in architectural design and training.

5. Discussion

This chapter reflects on the theoretical, practical and methodological implications of the findings, argues for their relevance more broadly with respect to generative artificial intelligence and its understandability. The thesis contributes to the existing theories by showing how a generative model, like Stable Diffusion, has trouble interpreting multi-word expressions, especially idioms and context dependents. Through measures like Intersection over Union (IoU) and instrument like Diffusion Attention Attribution Maps (DAAM), the study reveals how alignment happens at the semantic and visual levels of generative models. Further, it also highlights their shortcomings.

5.1 implications for the existing theory

Natural language processing has a hard time dealing with multi-word expressions or MWEs. Similarly, generative AI like Stable Diffusion also does. This study adds to research on AI interpretability. We apply the metric called intersection over union

(IoU) as a measure of how attention is aligned when developing models using MWEs. It provides us with insights into the semantic and visual relationship of MWEs.

New discoveries show that cross-attention mechanisms in generative AI systems are not perfect. Tang et al. (2022) also noted that these models struggle to interpret idiomatic phrases and multiword expressions (MWEs) as context requires. The previous study by Miletič and Walde (2024) had also noted these difficulties. In addition, the findings lend credence to the reasoning of Tang et al. (2022a) that linguistic difficulty frequently prevents generative models from attaining high semantic coherence across extended contexts. Following methods proposed in Chefer et al. (2021), we employ Diffusion Attention Attribution Maps (DAAM) to facilitate model understanding and model improvement through attention attribution. The study helps to distinguish between the holistic and component-level of IoU and thus extends the existing theory. It thereby offers a systematic way to ascertain whether a model views MWEs as a unified whole or apart from each other. This study fits in with recent research on the interpretability of artificial intelligence (AI), such as the Multimodal Automated Interpretability Agent (MAIA) developed by MIT researchers; MAIA uses vision-language models to automate interpretability tasks (MIT NEWS 2024). Additionally, it resonates with findings from studies on attention mechanisms in deep learning, which have become central to state-of-the-art AI models across various application domains (Springer, 2022). By situating these findings within the broader discourse on AI interpretability and MWE processing, this study contributes to a deeper understanding of the capabilities and limitations of generative AI models in handling complex linguistic constructs.

5.2 Implications in Practice

From a practical perspective, particularly in applications that rely on improved semantic alignment of MWEs, the principles outlined in this study provide valuable guidance for the development of generative artificial intelligence systems. Enhancing model designs to address the limitations identified in this work could significantly benefit sectors utilizing text-to-image generation, such as creative media, healthcare, and education. The high IoU scores achieved for MWEs like “crime scene investigators” and “thermonuclear fusion” underscore the importance of incorporating more literal MWEs into training datasets to enhance model performance.

However, this thesis also highlights the constraints of prompt engineering in addressing DAAM overlap issues. As Alaluf et al. (2022) point out, while rapid optimization techniques can improve the visual quality of outputs, they have minimal impact on the

distribution and alignment of model attention. Consequently, the findings emphasize the necessity of refining model architecture and attention mechanisms to achieve greater interpretability, rather than relying solely on fast optimization techniques.

5.3 Assignment's limits

This study presents interesting analysis, however its restrictions preclude the generalizability of its results. Though robust, the study's reliance on the Multiword Expression Complexity Word Identification dataset (MWE-CWI) (Ekochmar, n.d. 2020) would not fairly represent the linguistic diversity of MWEs among numerous languages and civilizations. This constraint affects the relevance of results to non-English MWEs or those of considerable cultural importance. Furthermore, this study was conducted exclusively on Stable Diffusion 2.1, which limits the generalizability of the findings to other text-to-image models. While prompt engineering was effective in circumventing the 77-token limit and improving the quality of generated images, it had minimal impact on DAAM-based attention alignment and, consequently, showed little influence on the IoU scores. Another significant limitation lies in the reliance on IoU as the primary evaluation metric. Although IoU provides a reliable measure of attention alignment, it does not adequately capture semantic alignment in instances where the generated visual outputs are abstract or ambiguous. This highlights the need for complementary evaluation metrics to address these gaps and provide a more comprehensive assessment of model performance. Complementary measurements include CLIP Score might provide a more complete knowledge of model performance, as indicated by Radford et al. (2021).

Ultimately, this work stresses the continuous challenges generative artificial intelligence models face in comprehending MWEs even if it offers theoretical and pragmatic gains to the field. Future study should investigate multi-lingual datasets, newer stable diffusion models like Stable Diffusion 3.5 series and other evaluation criteria like attention rollout or flow matching analysis in order to solve the discovered limitations and assure that generative artificial intelligence systems extend more inclusive and semantically strong.

6. Conclusion and perspectives

Focusing on how Stable Diffusion handles linguistically difficult inputs, this thesis investigated the interpretability problems presented by multi-word expressions (MWEs) in text-to-image generation. Using Intersection over Union (IoU) measures and

Diffusion Attention Attribution Maps (DAAM), the study offered both quantitative and qualitative analysis of how words aligned with visual outputs. With performance impacted by elements like linguistic difficulty, context dependency, and semantic compositionality, the results underlined notable variation in the model's capacity to depict MWEs.

The results help to clarify how generative models manage MWEs, therefore augmenting current research. Stable Diffusion showed expertise in understanding literal MWEs, including “brown dwarf” and “thermonuclear fusion,” but its limitations were visible with fixed phrase and context-specific language like “aqua vitae”, “Same sex marriage” This conflict emphasizes the requirement of improved attention mechanisms that give holistic semantic interpretation top priority over component-level, fractured processing.

In reality, the results have significance to improve generative artificial intelligence applications in many different fields. Semantic alignment is crucial in sectors such education, healthcare, and creative media, hence improved interpretability will help them as well. For example, by matching textual inputs with exact images, models that faithfully depict MWEs could help create linguistic relevant instructional materials.

The study did, however, also expose restrictions on the generalizability of its results. Although robust, depending alone on the Multiword Expression Complexity Word Identification dataset (MWE-CWI) (Ekochmar, n.d. 2020) could not fully represent the variety of MWEs in many languages and cultural settings. Furthermore, IoU measures do not fully explain semantic coherence or the quality of produced images even if they are useful for attention alignment. Complementary measurements like CLIP Score should be included into next studies to assess semantic alignment. Furthermore, it was discovered that prompt engineering, despite being essential for producing high-quality images, had very little effect on DAAM's attention overlap, indicating that architectural improvements are more important than input optimization.

Looking ahead, more research into multilingual and culturally complex datasets would help the field of generative artificial intelligence to improve the MWE representation. Semantic coherence of generative models could be improved by developments in cross-attention mechanisms and interpretability tools as DAAM. By addressing these problems, following studies can decrease the gap between textual and visual alignment, therefore enabling more accurate and sensitive artificial intelligence systems that better understand holistic meaning of the sentences rather than taking the meaning word by word.

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Attachment

Generated Images: The images generated during this study can be accessed via the following Google Drive link: [Generated Images](#).

Dataset: The annotated dataset used for this research and the IoU scores for all the related mwes and dataset images are available at this [link](#)

Declaration of Originality

I, Sayef Ali Khan, student registration number: 22929040, hereby confirm that I completed the submitted work independently and without the unauthorized assistance of third parties and without the use of undisclosed and unauthorized aids. This work has not been previously submitted in its current form or in a similar form to any other examination authorities and has not been accepted as part of an examination by any other examination authority.

Where the wording has been taken from other people's work or ideas, this has been properly acknowledged and referenced. This also applies to drawings, sketches, diagrams and sources from the Internet.

I am aware that the use of artificial intelligence is forbidden unless its use as an aid has been expressly permitted by the examiner. This applies to such programs that could process the task of the examination or parts thereof instead of me, for example chatbots (especially ChatGPT) and other methods suitable for this purpose.

Furthermore, I am aware that working with others in one room or by means of social or other media represents the unauthorized assistance of third parties within the above meaning, if group work is not expressly permitted. Each exchange of information with others during the examination, apart from examiners, persons authorized by them and invigilators, about the structure or contents of the examination or any other information such as sources is not permitted. The same applies to attempts to do so. Proofreading of the work by third parties with the aim of correcting linguistic deficiencies is permitted.

Any infringements of the above rules constitute fraud or attempted fraud and shall lead to the examination being graded "fail" ("nicht bestanden"). In the event of multiple or particularly drastic breaches of the rules, further sanctions may be imposed by the Examination Board (Prüfungsausschuss).

Erlangen, December 13, 2024

Place, Date

Signature